

QUESTION-25

DEPTH FIRST SEARCH

main.c	Run	Output
<pre>1 #include <stdio.h> 2 int graph[10][10], visited[10], n; 3 void dfs(int v) { 4 int i; 5 visited[v] = 1; 6 printf("%d ", v); 7 for(i = 0; i < n; i++) { 8 if(graph[v][i] && !visited[i]) 9 dfs(i); 10 } 11 } 12 int main() { 13 int i, j, start; 14 printf("Enter number of vertices: "); 15 scanf("%d", &n); 16 printf("Enter adjacency matrix:\n"); 17 for(i = 0; i < n; i++) 18 for(j = 0; j < n; j++) 19 scanf("%d", &graph[i][j]); 20 printf("Enter starting vertex: ");</pre>		<pre>Enter number of vertices: 2 Enter adjacency matrix: 2 2 3 1 3 5 Enter starting vertex: DFS Traversal: 3 === Code Execution Successful ===</pre>

main.c	Run	Output
<pre>6 printf("%d ", v); 7 for(i = 0; i < n; i++) { 8 if(graph[v][i] && !visited[i]) 9 dfs(i); 10 } 11 } 12 int main() { 13 int i, j, start; 14 printf("Enter number of vertices: "); 15 scanf("%d", &n); 16 printf("Enter adjacency matrix:\n"); 17 for(i = 0; i < n; i++) 18 for(j = 0; j < n; j++) 19 scanf("%d", &graph[i][j]); 20 printf("Enter starting vertex: "); 21 scanf("%d", &start); 22 printf("DFS Traversal: "); 23 dfs(start); 24 return 0; 25 }</pre>		<pre>Enter number of vertices: 2 Enter adjacency matrix: 2 2 3 1 3 5 Enter starting vertex: DFS Traversal: 3 === Code Execution Successful ===</pre>